TraceBack — AI Engineering Log

Nov 6, 2026 · @Soumyaranjan

Across the project the team logged 41 significant AI-assisted decisions: 12 accepted as proposed, 19 modified and 10 rejected. Deliverable 3 adds 15 entries (27–41); in each, the AI's output was checked by running tests, profilers or audits before it was kept, and 10 of the 15 were changed or rejected after that check.

TraceBack is the IIT Kanpur Lost & Found portal. Two kinds of AI are involved. Inside the product, Groq (`openai/gpt-oss-20b`) runs the matching agent and Google Gemini (`gemini-2.5-flash`) is its fallback and describes photos, both on free-tier API keys with a cap of 500 calls a day. During development, the team used an LLM coding assistant (Claude, by Anthropic) to write code, tests, test scripts and report drafts. This log covers the second kind: what we asked, what the AI produced, what was wrong or risky, and what we kept.

How AI output was checked

- **Nothing merged on the AI's word.** Every AI-written change went through the full pytest suite, and the behaviour it claimed was measured: load with the load generator and py-spy, races with the concurrency suite, security with the attack script and the audits, recovery with the failure suite.
- **Tests first for fixes.** For each defect, a test or scenario that reproduces it was run before the fix and after it.
- **Numbers from files, not memory.** Every figure in the reports was taken from a result file in `testing/results/` or a fetched source, then cross-checked between reports.
- **The model stays at the edge.** The product rule from Deliverable 2 still holds: the LLM proposes rankings and reasons; code owns queries, dates, thresholds, emails and every status change.

Deliverable 3 entries (27–41)

Author of entries 27–41: Soumyaranjan Sahu, working with the coding assistant on testing, deployment and hardening.

#	Prompt (summary)	AI output	Our evaluation, issues found	Change made, final artifact	Decision
27	Find out why the API stops answering at 25 users	A load generator and a py-spy profile showing every worker thread blocked on the database pool; proposed capping the requests in flight	Confirmed by re-running every load step after each change: 40 threads waited on 15 connections while their holders waited for a thread; enlarging the pool alone would only move the limit	`core/capacity.py` caps requests in flight at 30 per process, with the pool sized to match; 0 errors at 100 users (D-01)	Accepted
28	Stop uploads from blocking the event loop	Make the upload endpoints plain `def` so they run in threads	Correct for speed, but concurrency test CON-10 then let 6 parallel uploads pass the 3-photo limit	Item row locked while photos are counted; CON-10 passes 40 of 40 (D-25)	Modified
29	Write tests for simultaneous requests	Script that releases requests together with a thread barrier, one connection each, repeated per race	Each race repeated 10 and then 30 times on fresh items, so a race had many chances to overlap; results read from the database, not just the HTTP answers	`testing/concurrency_tests.py`, 14 checks; it found D-02 to D-07	Accepted
30	Fix the claim and custody races	Lock the item row, then the claim row, in every action, and re-read before checking the rules	Checked the lock order for deadlocks (none in the PostgreSQL log over 40 trials) and that the existing 189 tests still pass	`claims._lock`, `custody._lock`; CON-04 to CON-09 pass	Accepted
31	Tidy the new test files	Ran an auto-fixer over the tests folder	It also removed deliberate `# noqa: E402` comments from `conftest.py`, which must import after setting environment variables	Fixer's change to `conftest.py` discarded; file rebuilt from the original plus the new fixture	Rejected

#	Prompt (summary)	AI output	Our evaluation, issues found	Change made, final artifact	Decision
32	Raise coverage with meaningful tests, no exclusions	235 tests for the CLI, worker, AI clients, email and API edges	Some test data broke database rules (found items without a drop-off office, an invalid agent outcome, a second main office): the data was wrong, not the app	Test data fixed to respect every constraint; 98% line coverage with no excluded file	Modified
33	Refuse decompression bombs	Read the image size and refuse anything over 60 megapixels	Broke existing tests that upload placeholder bytes Pillow can't read	Unreadable headers accepted as before (they get no thumbnail); real bombs refused (D-08)	Modified
34	Review the system-test screenshots for defects	Reported that the sidebar background stops short on long pages	A full-page screenshot draws a sticky sidebar only once; in a real browser, scrolled, the sidebar stays full height	Finding withdrawn; no change to the app	Rejected
35	Upgrade Pillow to clear 39 advisories	Pin `pillow==13.0.0`	`pip` reported that fastembed 0.9.0 requires Pillow below 13, which would have broken the Docker build on the server	fastembed 0.9.1 as well; full dependency resolution checked; tests and audits clean (D-12)	Modified
36	Remove GPS data from uploaded photos	Clean JPEGs segment by segment instead of re-encoding them, keeping only the rotation flag	Verified pixel-identical output, rotation kept, XMP and trailing second images removed, damaged files refused	`storage.strip_metadata`; ADV-17 stopped (D-13)	Accepted
37	Fix the "PLACEHOLDER" office text in emails	Option considered: fill in plausible directions for each office	Invented directions would send students to the wrong place	No seeded directions; emails omit missing ones; `set-office` command for the real data (D-16)	Rejected
38	Write failure and recovery tests	Script that stops the database, kills processes and swaps in a fake mail server	First run gave false results: killing uvicorn's parent left its worker processes serving, and two endpoint paths were wrong	Processes killed by process group, paths fixed; the corrected run found 4 real defects	Modified

#	Prompt (summary)	AI output	Our evaluation, issues found	Change made, final artifact	Decision
39	Fix the 4 recovery defects	503 handler, report and AI job in one commit, interrupted jobs re-run with a `started_at` column, unsent codes removed	Each verified by a new test that fails on the old code (9 of 10 failed there) and by the full failure suite	`9fd794f`; FR-01 to FR-10 all pass (D-19 to D-22)	Accepted
40	Estimate the monthly AWS cost	US$16.35 a month for the instance	That is the Mumbai price; the server runs in Sydney, where the same instance costs US$19.27	Costing uses fetched Sydney prices: US$24.84 a month in all	Modified
41	Draft the test reports from the result files	Report text and charts	Cross-checks found figures from runs with different settings side by side (1 and 2 API processes) and test counts out of date after later fixes	Each chart labelled with its run; all counts re-taken from the final result files	Modified

Deliverable 2 entries (1–26), in brief

The full entries are in the Deliverable 2 report (section 7).

#	Member	Topic	Decision
1	Rohit Yadav	Semantic top-k matching design	Accepted
2	Rohit Yadav	LLM agent on every report, changed to ambiguous-only with a daily limit	Modified
3	Rohit Yadav	Free LLM providers (a provider listed as free needed a top-up)	Rejected
4	Rohit Yadav	Agent’s match reasons (a hallucinated date gap)	Modified
5	Rohit Yadav	Sending item photos to Gemini (ID cards excluded)	Modified
6	Rohit Yadav	Agent tool to find any item’s owner	Rejected
7	Team	Roll number in the AI description	Rejected
8	Rohit Yadav	Offline OCR engine for ID cards	Accepted
9	Rohit Yadav	Who to email from a found ID card	Modified

#	Member	Topic	Decision
10	Rohit Yadav	Pinning the OpenAI SDK	Modified
11	Rohit Yadav	Whether a Gemini subscription raises API limits	Accepted
12	Rohit Yadav	Handling Groq rate limits	Rejected
13	Soumyaranjan Sahu	Real, branded emails with an email log	Accepted
14	Team	The TraceBack logo	Accepted
15	Rohit Yadav	Splitting the code into story PRs	Modified
16	Soumyaranjan Sahu	Diagnosing “the owner got no email”	Accepted
17	Ankit Agarwal	Storing sign-in codes and limiting guesses	Modified
18	Ankit Agarwal	Keeping users signed in (JWT in localStorage)	Rejected
19	Ankit Kumar Dubey	Two guards approving the same item	Modified
20	Ankit Kumar Dubey	Claim state transitions	Modified
21	Bhanu Pratap	Text search	Rejected
22	Bhanu Pratap	Validating uploaded photos	Modified
23	Dev Sachan	Modelling custody history	Modified
24	Dev Sachan	Staff dashboard stages	Accepted
25	Soumyaranjan Sahu	Reminder emails from the worker	Modified
26	Soumyaranjan Sahu	Inviting staff by email	Rejected

Lessons from Deliverable 3

- **An AI’s test is still a test to check.** The first failure-test run reported passes and failures that were the script’s own mistakes (entry 38); checking the tests found them before they reached a report.
- **Speed fixes create races.** Moving uploads to threads (entry 28) broke a limit that only a concurrency test could see; performance and concurrency suites belong together.

- **Plausible is not correct.** A price for the wrong region (40), a defect that was a screenshot artefact (34) and a dependency conflict hidden behind a one-line upgrade (35) all looked right until measured.
- **Prefer leaving data out to inventing it.** Missing office directions are better than made-up ones (37).